INCH-POUND

MIL-O-55310/6D
15 March 1994
SUPERSEDING
MIL-O-55310/6C
4 October 1984

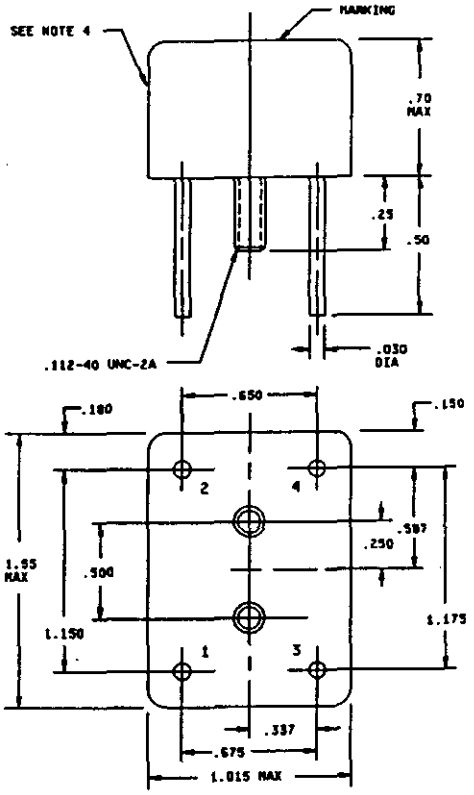
MILITARY SPECIFICATION SHEET

OSCILLATOR, CRYSTAL CONTROLLED, TYPE 1 (CRYSTAL OSCILLATOR (XO)), 15.7 kHz THROUGH 10.0 MHz, SOLDER SEAL, SQUARE WAVE, TTL

Inactive for new design

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-O-55310



Pin number	Function
1	RF output
2	B+
3	B- (common)
4	Case

Inches	mm
.030	0.76
.150	3.81
.180	4.57
.250	6.35
.337	8.56
.500	12.70
.587	14.91
.650	16.51
.675	17.14
.70	17.8
1.015	25.78
1.150	29.21
1.175	29.84
1.55	39.4

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm) for three place decimals and ±.02 (0.5 mm) for two place decimals.
4. Pin number and function to be displayed on side of oscillator.

FIGURE 1. Dimensions and configuration.

AMSC N/A

FSC 5955

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REQUIREMENTS:

Design and construction: See figure 1.

Mounting: See figure 1.

Terminals: See figure 1.

Seal: Solder (nonhermetic).

Weight: 3 ounces maximum.

Construction technology: Discrete construction.

Components: In accordance with MIL-STD-454.

Calibration: Manufacturer calibrated.

Screening: In accordance with MIL-O-55310, class B.

Temperature:

Operating: -55°C to +85°C.

Storage: -62°C to +125°C.

Oscillator load: 1 to 10 standard TTL loads. A TTL unit load is defined as: 1.6 mA sink, 0.04 mA source and 2 pF capacitance.

Output waveform: Symmetrical square wave, TTL compatible.

Supply voltage: +5 V dc ±0.25 V dc (+6 V dc shall not damage oscillator).

Input current: 105 mA maximum.

Output frequency: Range shall be 15.7 kHz through 10.0 MHz; frequency as designated at time of acquisition.

Output voltage:

Logic 1: 2.4 volts minimum at 400 µA maximum.

Logic 0: 0.4 volt maximum at 16 mA maximum.

Rise and fall times: 200 ns to 25 ns, frequency dependent, designated at time of acquisition.

Duty cycle: 45 to 55 percent duty cycle at 1.4 V.

Initial frequency-temperature accuracy (one-half temperature cycle): ±50 ppm from -55°C to +85°C. Measurements taken at 10 equispaced increments over the specified operating temperature range.

Frequency-voltage tolerance: ±2.5 ppm maximum for a ±0.25 volt change in supply voltage. Measurements taken at reference temperature and operating temperature range end points.

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Frequency aging: Aging measurements shall be taken at +70°C ±0.2°C.

±0.25 ppm per 30 days, maximum.

±0.5 ppm per 90 days, maximum. 1/

±5 ppm per year maximum. 1/

Terminal strength: MIL-STD-202, method 211, test conditions A and B.

Applied force: 5 pounds.

Bends: Five at 45° each.

Frequency-environmental tolerance: Not applicable.

Vibration, sinusoidal: In accordance with MIL-O-55310 and MIL-STD-202, method 204.

Nonoperating: Test condition B.

Operating: Test condition C.

Ambient pressure:

Nonoperating: In accordance with MIL-O-55310.

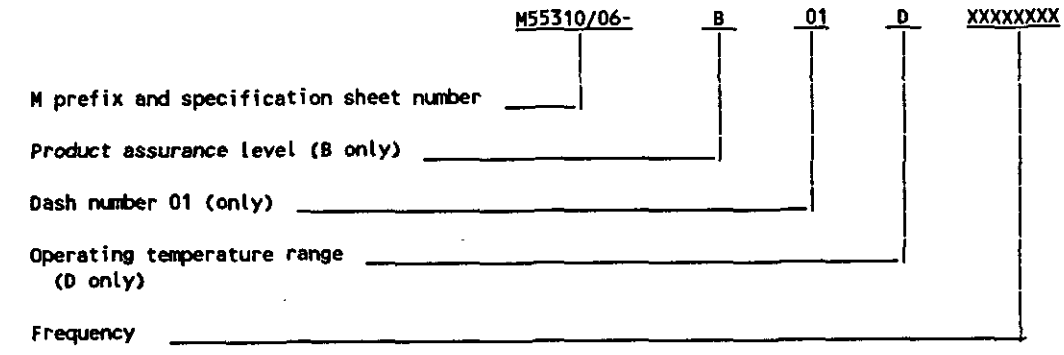
Operating: MIL-STD-202, method 105, test condition B.

QUALITY ASSURANCE PROVISIONS

Group C inspection: Not applicable.

Part or Identifying Number (PIN): Consists of "M" prefix followed by specification sheet number, a dash and code alphas, and numeric number. See example:

EXAMPLE



Revision letters are not used to denote changes due to the extensiveness of the changes.

1/ This is a performance requirement of the oscillator. Support data shall be presented showing that this requirement shall be met.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 19, 99
DLA - ES

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5955-0677-03)

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